

Smart pointers

Implement class `House` for storing housing information.

The class must use `std::string` for storing the address of the house, a double for the size (area in m²) of the house and the price of the house in euros. All data members must be private.

Class must define a constructor that takes the house information as parameters but no default constructor (constructor that takes no parameters).

The class must have the following operators:

- Output operator for printing the housing information with minimal formatting. **No newlines may be printed by the operator.**
- Less than operator that compares two houses by price/m²

Class must define the following member functions:

- `setAddress` for setting the address of the house
- `getAddress` for getting the address of the house
- `setArea` for setting the area of the house
- `setPrice` for setting the price of the house

Implement class `Apartment` for storing information about rental apartments

The class must contain a `shared_ptr` to `House` (the class described above), apartment number, size of the apartment in m² and the monthly rent. All data members must be private.

There can be multiple apartments in a house and each apartment has a smart pointer to the house it is in, and the pointer can be used to get the address of the apartment.

The apartment class must implement less than operator that compares apartments by monthly rent.

The apartment class must have a member function that prints the apartment information:

- Address and apartment nr
- Size of the apartment
- The monthly rent

Implement a program that creates two houses and a vector of Apartments

Program adds three apartments of the first house and four apartments of the second house to the vector. Then program sorts the vector and prints the apartment information on separate lines.